



Renewable Energy

Power from sources that naturally replenish

043

Renewable Energy comes from naturally replenishing sources like sunlight, wind, and water, offering a cleaner alternative to fossil fuels for electricity and heat generation.

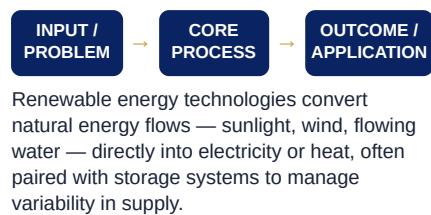
● DEFINITION

Energy derived from natural sources that are replenished at a faster rate than they are consumed, such as sunlight, wind, and water.

● WHY IT MATTERS

Renewable energy reduces dependence on fossil fuels and greenhouse gas emissions, playing a central role in addressing climate change and energy security.

● WORKING PRINCIPLE



● QUICK FACTS

- ▶ Solar panel costs have dropped by roughly 90% since 2010
- ▶ Renewable sources now supply around 30% of global electricity generation

● KEY TERMS

Solar power Wind power
Energy storage Grid integration

● CORE CONCEPTS

- ▶ Solar and wind power generation
- ▶ Energy storage integration
- ▶ Grid integration
- ▶ Levelized cost of energy

● APPLICATIONS

- ▶ Solar and wind power plants
- ▶ Hydroelectric power generation
- ▶ Renewable-powered microgrids
- ▶ Electric vehicle charging infrastructure

● ADVANTAGES

- ▶ Reduces greenhouse gas emissions
- ▶ Increasingly cost-competitive with fossil fuels
- ▶ Enhances energy independence

CAREER OPPORTUNITIES

Renewable energy engineer, Solar/wind technician, Energy policy analyst, Grid integration specialist

RESEARCH AREAS

Energy storage technology, Grid integration solutions, Next-generation solar cells

● REAL-LIFE EXAMPLES

- ▶ Rooftop solar panels
- ▶ Wind farms generating grid electricity
- ▶ Hydroelectric dams

● LIMITATIONS

- ▶ Solar and wind output varies with weather and time of day
- ▶ Requires energy storage or backup for reliability
- ▶ Large installations require significant land or space

INDUSTRY APPLICATIONS

Utilities, Construction, Automotive (EVs), Government infrastructure

FUTURE SCOPE

Rapidly falling costs and strong policy support are driving continued exponential growth in global renewable energy capacity.

● CURRENT TRENDS

Solar and wind power costs have fallen so dramatically they are now often the cheapest new electricity sources in many regions.

● SUMMARY

Renewable energy converts natural sources like sunlight and wind into clean electricity, now among the fastest-growing energy sectors globally.

